

# Exercise 2 – Solution Proposal

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Electric Double Layer Modeling: Gouy–Chapman

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## Introduction

In this document, we investigate the Gouy–Chapman model for the electric double layer. The questions and notation are taken from the problem sheet [1]. The Gouy–Chapman model describes the electrostatic potential and the ion concentration profiles in a diffuse double layer.

### Equations: Gouy–Chapman equation

The one-dimensional Gouy–Chapman equation for the electric potential  $\phi(x)$  is

$$\frac{d^2\phi}{dx^2} = -\frac{2e_0N_Ac_b}{\varepsilon_r\varepsilon_0} \sinh\left(\frac{e_0\phi}{k_BT}\right), \quad x \in ]0, \infty[ \quad (1)$$

with boundary conditions

$$\phi(x=0) = \phi_M - \phi_{pzc}, \quad (2)$$

$$\lim_{x \rightarrow \infty} \phi(x) = 0. \quad (3)$$

Here,  $\phi_M - \phi_{pzc}$  is the applied surface potential relative to the potential of zero charge.

### Equations: Ion concentration profiles

The anion and cation concentration profiles are given by

$$c_-(x) = c_b \exp\left(\frac{e_0\phi(x)}{k_BT}\right) \quad (4)$$

$$c_+(x) = c_b \exp\left(-\frac{e_0\phi(x)}{k_BT}\right) \quad (5)$$

where  $c_b$  is the bulk concentration.

### Equations: Exact Gouy–Chapman solution

The analytical solution of the Gouy–Chapman equation is

$$\phi^{(GC)}(x) = \frac{4k_BT}{e_0} \operatorname{arctanh}\left[\tanh\left(\frac{e_0(\phi_M - \phi_{pzc})}{4k_BT}\right) \exp(-\lambda_D^{-1}x)\right] \quad (6)$$

with the inverse Debye length

$$\lambda_D^{-1} = \left(\frac{2e_0^2N_Ac_b}{\varepsilon_0\varepsilon_r k_BT}\right)^{1/2} \quad (7)$$

### Equations: Surface charge density and double-layer capacitance

The surface free charge density is defined by

$$\sigma_{\text{free}} = -\varepsilon_0\varepsilon_r \left.\frac{d\phi}{dx}\right|_{x=0^+}. \quad (8)$$

For the Gouy–Chapman model, this gives

$$\sigma_{\text{free}}^{(GC)}(\phi_M - \phi_{pzc}) = \sqrt{8k_BT\varepsilon_0\varepsilon_r N_A c_b} \sinh\left(\frac{e_0(\phi_M - \phi_{pzc})}{2k_BT}\right) \quad (9)$$

The corresponding double-layer capacitance is

$$C_{\text{dl}} = \frac{d\sigma_{\text{free}}}{d(\phi_{\text{M}} - \phi_{\text{pzc}})}, \quad (10)$$

and therefore

$$C_{\text{dl}}^{(\text{GC})}(\phi_{\text{M}} - \phi_{\text{pzc}}) = \varepsilon_0 \varepsilon_r \lambda_D^{-1} \cosh\left(\frac{e_0(\phi_{\text{M}} - \phi_{\text{pzc}})}{2k_{\text{B}}T}\right). \quad (11)$$

Table 1: Fundamental constants

| Symbol          | Value                   | Unit              |
|-----------------|-------------------------|-------------------|
| $N_{\text{A}}$  | $6.022 \times 10^{23}$  | $\text{mol}^{-1}$ |
| $k_{\text{B}}$  | $1.381 \times 10^{-23}$ | $\text{J K}^{-1}$ |
| $e_0$           | $1.602 \times 10^{-19}$ | C                 |
| $\varepsilon_0$ | $8.854 \times 10^{-12}$ | $\text{F m}^{-1}$ |

\* Values are given in SI units.

Table 2: Model parameters and variables in the Gouy–Chapman model

| Symbol                                | Unit                | Meaning                                                                    |
|---------------------------------------|---------------------|----------------------------------------------------------------------------|
| $\phi_{\text{M}} - \phi_{\text{pzc}}$ | V                   | Applied metal potential relative to the potential of zero charge.          |
| $T$                                   | K                   | Temperature.                                                               |
| $\varepsilon_r$                       | —                   | Relative permittivity of the solvent.                                      |
| $c_b$                                 | $\text{mol m}^{-3}$ | Bulk ion concentration.                                                    |
| $\lambda_D$                           | m                   | Debye length, i.e. the characteristic screening length of the electrolyte. |
| $x$                                   | m                   | Distance from the metal surface into the electrolyte.                      |

# 1 Model setup: Gouy–Chapman

## Question: 1 Gouy–Chapman model setup

For the Gouy–Chapman model, please answer the following questions:

- 1.1) How does the electric potential change with varying  $(\phi_M - \phi_{pzc})$ ?
- 1.2) At which  $(\phi_M - \phi_{pzc})$  does  $\phi$  flip?
- 1.3) How do the concentrations change with varying  $(\phi_M - \phi_{pzc})$ , respectively?
- 1.4) At which  $(\phi_M - \phi_{pzc})$  do the concentrations flip?
- 1.5) Comment on the concentration densities near the metal surface.

## Solution: 1 Gouy–Chapman model setup

For convenience, we introduce the surface potential relative to the potential of zero charge,

$$\Delta\phi = \phi_M - \phi_{pzc}. \quad (12)$$

### 1.1) Electric potential as a function of $\Delta\phi$

From eq. (6), the electric potential is

$$\phi^{(GC)}(x) = \frac{4k_B T}{e_0} \operatorname{arctanh} \left[ \tanh \left( \frac{e_0 \Delta\phi}{4k_B T} \right) \exp(-\lambda_D^{-1} x) \right]. \quad (13)$$

At the metal surface, the boundary condition gives

$$\phi(0) = \Delta\phi. \quad (14)$$

Therefore, increasing  $\Delta\phi$  increases the value of the electric potential at the surface. Furthermore, for all finite values of  $\Delta\phi$ , the potential decays into the electrolyte,

$$\lim_{x \rightarrow \infty} \phi(x) = 0. \quad (15)$$

Thus, the Gouy–Chapman potential is largest in magnitude close to the metal surface and approaches zero in the bulk electrolyte.

Hence,

$$\boxed{\Delta\phi > 0 \implies \phi(x) > 0}$$

and

$$\boxed{\Delta\phi < 0 \implies \phi(x) < 0}$$

for  $x$  close to the surface. In both cases,  $\phi(x)$  approaches zero as  $x \rightarrow \infty$ .

### 1.2) Potential flip

The potential changes sign when the applied surface potential changes sign. This occurs at the potential of zero charge:

$$\Delta\phi = 0. \quad (16)$$

Since  $\Delta\phi = \phi_M - \phi_{pzc}$ , this is equivalent to

$$\phi_M = \phi_{pzc}. \quad (17)$$

Therefore, the electric potential flips at

$$\boxed{\phi_M - \phi_{pzc} = 0}.$$

This potential flip is shown in Figure 1. For  $\phi_M - \phi_{pzc} < 0$ , the electric potential is negative; for  $\phi_M - \phi_{pzc} > 0$ , it is positive. At  $\phi_M - \phi_{pzc} = 0$ , the electric potential is zero everywhere.

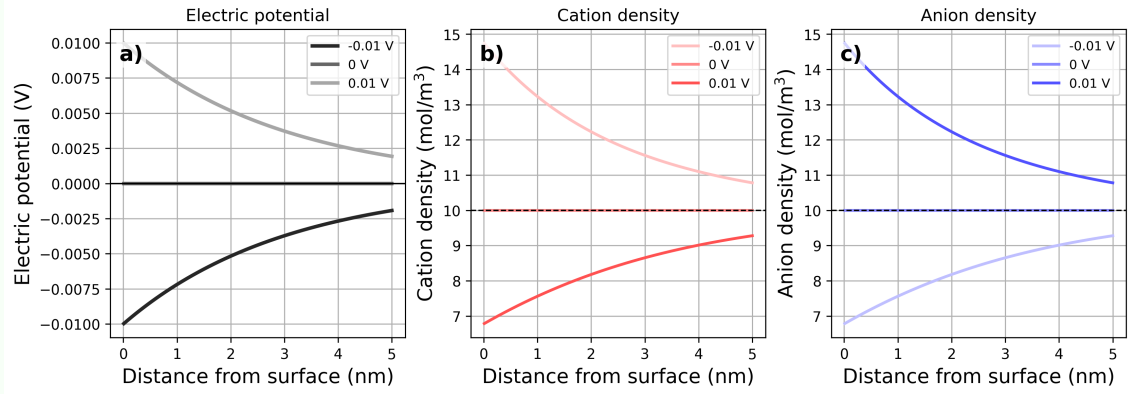


Figure 1: Gouy-Chapman potential and ion concentration profiles for negative, zero, and positive applied surface potential. Panel a) shows the electric potential, panel b) shows the cation density, and panel c) shows the anion density. The electric potential changes sign at  $\phi_M - \phi_{pzc} = 0$ . At the same point, the ion concentration profiles also flip.

### 1.3) Concentrations as a function of $\Delta\phi$

The concentration profiles are

$$c_-(x) = c_b \exp\left(\frac{e_0\phi(x)}{k_B T}\right) \quad (18a)$$

$$c_+(x) = c_b \exp\left(-\frac{e_0\phi(x)}{k_B T}\right) \quad (18b)$$

If the surface potential is positive, i.e.

$$\Delta\phi > 0 \quad (19)$$

then  $\phi(x) > 0$  close to the surface. Therefore,

$$c_-(x) = c_b \exp\left(\frac{e_0\phi(x)}{k_B T}\right) > c_b \quad (20)$$

and

$$c_+(x) = c_b \exp\left(-\frac{e_0\phi(x)}{k_B T}\right) < c_b \quad (21)$$

Thus, for a positive surface potential, anions accumulate near the surface and cations are depleted.

If the surface potential is negative, i.e.

$$\Delta\phi < 0 \quad (22)$$

then  $\phi(x) < 0$  close to the surface. Therefore,

$$c_-(x) < c_b \quad (23)$$

and

$$c_+(x) > c_b \quad (24)$$

Thus, for a negative surface potential, cations accumulate near the surface and anions are depleted.

In summary,

$$\Delta\phi > 0 : \quad c_- > c_b, \quad c_+ < c_b$$

and

$$\Delta\phi < 0 : \quad c_- < c_b, \quad c_+ > c_b$$

#### 1.4) Concentration flip

The concentrations flip when the electric potential changes sign. Since the electric potential flips at

$$\Delta\phi = 0, \quad (25)$$

the concentrations also flip at

$$\phi_M - \phi_{pzc} = 0$$

At this point,

$$\phi(x) = 0 \quad (26)$$

and therefore

$$c_-(x) = c_b, \quad c_+(x) = c_b \quad (27)$$

Thus, at the potential of zero charge, neither anions nor cations are preferentially accumulated at the surface.

This concentration flip is also visible in Figure 1b,c. In Figure 1b, the cation density is enhanced for negative surface potential and depleted for positive surface potential. In Figure 1c, the anion density shows the opposite behavior: it is depleted for negative surface potential and enhanced for positive surface potential. For  $\phi_M - \phi_{pzc} = 0$ , both ion densities remain equal to the bulk concentration  $c_b$ .

### 1.5) Concentration densities near the metal surface

Near the metal surface, the potential has its largest magnitude. Since the concentration profiles depend exponentially on the electric potential, small changes in  $\phi(x)$  can produce large changes in the ion concentrations.

For a positive surface potential,

$$\phi(0) > 0 \quad (28)$$

and therefore the anion density close to the metal surface becomes large:

$$c_-(0) = c_b \exp\left(\frac{e_0 \Delta \phi}{k_B T}\right) \quad (29)$$

At the same time, the cation density is strongly reduced:

$$c_+(0) = c_b \exp\left(-\frac{e_0 \Delta \phi}{k_B T}\right) \quad (30)$$

For a negative surface potential, the roles are reversed. Cations accumulate and anions are depleted.

This is one of the important limitations of the Gouy–Chapman model: the ions are treated as point-like particles. Therefore, the model can predict unrealistically large ion concentrations close to the metal surface, especially for large values of  $|\phi_M - \phi_{pzc}|$ . Hence,

The Gouy–Chapman model predicts strong counter-ion accumulation near the surface.

However,

for large  $|\Delta \phi|$ , the predicted near-surface densities can become unphysical.

## 2 Surface charge and capacitance

### Question: $\sigma_{\text{free}}$ and $C_{\text{dl}}$ : Gouy–Chapman

For the Gouy–Chapman surface charge density and double-layer capacitance, please answer the following questions:

- 2.1) Show that if  $\phi_{\text{M}} = \phi_{\text{pzc}}$ , then  $\sigma_{\text{free}}^{(\text{GC})} = 0$ . Hint:  $\sinh(0) = 0$ .
- 2.2) Derive  $C_{\text{dl}}^{(\text{GC})}(0)$ . Hint:  $\cosh(0) = 1$ .
- 2.3) Using the provided code and/or the figure below, assess to which extent the GC model agrees with the experimental data.

### Solution: $\sigma_{\text{free}}$ and $C_{\text{dl}}$ : Gouy–Chapman

#### 2.1) Surface charge density at the potential of zero charge

The Gouy–Chapman surface free charge density is

$$\sigma_{\text{free}}^{(\text{GC})}(\phi_{\text{M}} - \phi_{\text{pzc}}) = \sqrt{8k_{\text{B}}T\varepsilon_0\varepsilon_r N_{\text{A}} c_{\text{b}}} \sinh\left(\frac{e_0(\phi_{\text{M}} - \phi_{\text{pzc}})}{2k_{\text{B}}T}\right) \quad (31)$$

If

$$\phi_{\text{M}} = \phi_{\text{pzc}}, \quad (32)$$

then

$$\phi_{\text{M}} - \phi_{\text{pzc}} = 0 \quad (33)$$

Inserting this into the Gouy–Chapman expression gives

$$\sigma_{\text{free}}^{(\text{GC})}(0) = \sqrt{8k_{\text{B}}T\varepsilon_0\varepsilon_r N_{\text{A}} c_{\text{b}}} \sinh\left(\frac{e_0 \cdot 0}{2k_{\text{B}}T}\right) = \sqrt{8k_{\text{B}}T\varepsilon_0\varepsilon_r N_{\text{A}} c_{\text{b}}} \sinh(0) \quad (34)$$

Using

$$\sinh(0) = 0 \quad (35)$$

we obtain

$$\sigma_{\text{free}}^{(\text{GC})}(0) = 0 \quad (36)$$

Therefore,

$$\boxed{\phi_{\text{M}} = \phi_{\text{pzc}} \implies \sigma_{\text{free}}^{(\text{GC})} = 0}$$

#### 2.2) Capacitance at the potential of zero charge

The Gouy–Chapman double-layer capacitance is

$$C_{\text{dl}}^{(\text{GC})}(\phi_{\text{M}} - \phi_{\text{pzc}}) = \varepsilon_0\varepsilon_r\lambda_D^{-1} \cosh\left(\frac{e_0(\phi_{\text{M}} - \phi_{\text{pzc}})}{2k_{\text{B}}T}\right) \quad (37)$$

At the potential of zero charge,

$$\phi_{\text{M}} - \phi_{\text{pzc}} = 0 \quad (38)$$



Therefore,

$$C_{\text{dl}}^{(\text{GC})}(0) = \varepsilon_0 \varepsilon_r \lambda_D^{-1} \cosh\left(\frac{e_0 \cdot 0}{2k_B T}\right) \quad (39)$$

$$= \varepsilon_0 \varepsilon_r \lambda_D^{-1} \cosh(0) \quad (40)$$

Using

$$\cosh(0) = 1 \quad (41)$$

we obtain

$$C_{\text{dl}}^{(\text{GC})}(0) = \varepsilon_0 \varepsilon_r \lambda_D^{-1} \quad (42)$$

Thus,

$$\boxed{C_{\text{dl}}^{(\text{GC})}(0) = \varepsilon_0 \varepsilon_r \lambda_D^{-1}}$$

Using the expression for the inverse Debye length,

$$\lambda_D^{-1} = \left( \frac{2e_0^2 N_A c_b}{\varepsilon_0 \varepsilon_r k_B T} \right)^{1/2} \quad (43)$$

we can also write

$$C_{\text{dl}}^{(\text{GC})}(0) = \varepsilon_0 \varepsilon_r \left( \frac{2e_0^2 N_A c_b}{\varepsilon_0 \varepsilon_r k_B T} \right)^{1/2} \quad (44)$$

Equivalently,

$$\boxed{C_{\text{dl}}^{(\text{GC})}(0) = \left( \frac{2e_0^2 N_A c_b \varepsilon_0 \varepsilon_r}{k_B T} \right)^{1/2}}$$

### 2.3) Comparison with the experimental data

The Gouy–Chapman capacitance is

$$C_{\text{dl}}^{(\text{GC})}(\Delta\phi) = \varepsilon_0 \varepsilon_r \lambda_D^{-1} \cosh\left(\frac{e_0 \Delta\phi}{2k_B T}\right), \quad \Delta\phi = \phi_M - \phi_{\text{pzc}} \quad (45)$$

Since the hyperbolic cosine has a minimum at zero,

$$\cosh(0) = 1 \quad (46)$$

the Gouy–Chapman model predicts a capacitance minimum at the potential of zero charge. For positive and negative deviations from the potential of zero charge, the capacitance increases symmetrically.

Therefore, the Gouy–Chapman prediction has a characteristic “U-shape”:

$$\boxed{C_{\text{dl}}^{(\text{GC})} \text{ is minimal at } \phi_M = \phi_{\text{pzc}} \text{ and increases for larger } |\Delta\phi|}$$

This behavior is compared with the experimental Valette data in Figure 2.

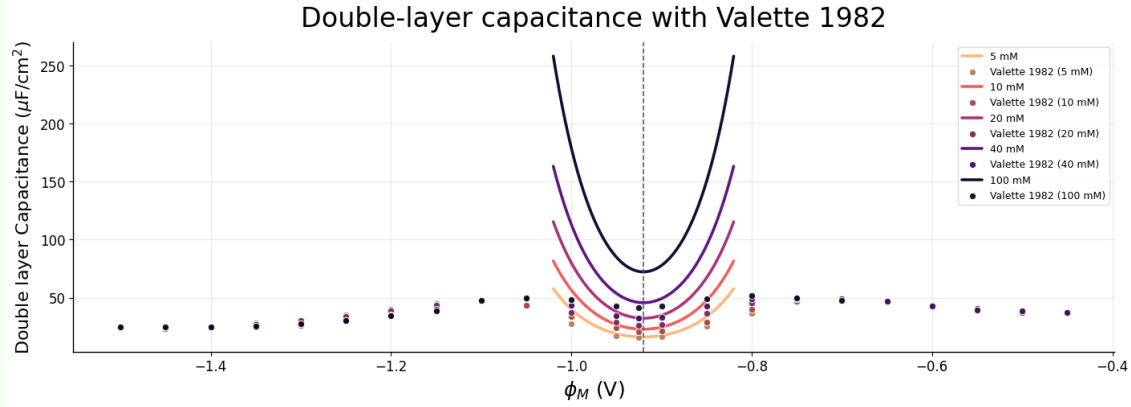


Figure 2: Comparison between the Gouy–Chapman double-layer capacitance and the experimental capacitance data of Valette. The Gouy–Chapman model predicts a symmetric U-shaped capacitance curve, whereas the experimental data are much flatter over the same potential range.

As shown in Figure 2, the experimental Valette data do not show the same strong symmetric increase over the whole potential range. In particular, the experimental capacitance is much flatter than the ideal Gouy–Chapman prediction for many potentials. The Gouy–Chapman model can reproduce the qualitative idea that the capacitance has a minimum near the potential of zero charge, but it overestimates the capacitance strongly away from the minimum.

Hence, the agreement is limited:

GC captures the qualitative minimum near the PZC, but fails quantitatively away from it

The main reason is that the Gouy–Chapman model only describes a diffuse layer of point ions. It neglects finite ion size, specific adsorption, and solvent structure. These effects are important for real metal–electrolyte interfaces and therefore for comparison with experimental capacitance data.

### 3 Overall model assessment

#### Question: 3 Overall model assessment

Please answer the following questions:

- 3.1) What are the shortcomings of the Gouy–Chapman model?
- 3.2) What are the strengths of the Gouy–Chapman model?

#### Solution: 3 Overall model assessment

##### 3.1) Shortcomings of the Gouy–Chapman model

The main shortcomings of the Gouy–Chapman model are:

- **Point-ion approximation:** Ions are treated as point charges. Therefore, the model does not account for the finite size of ions.
- **No maximum ion concentration:** Since ions have no volume in the model, there is no upper limit for the ion concentration close to the surface.
- **Unphysical near-surface concentrations:** The ion concentrations depend exponentially on the electrostatic potential. For large surface potentials, this can lead to unrealistically large counter-ion concentrations near the metal surface.
- **No finite distance of closest approach:** In reality, ions cannot approach the metal surface arbitrarily closely because of their finite size and solvation shell. The Gouy–Chapman model does not include this effect.
- **Constant continuum permittivity:** The solvent is treated as a continuum with constant relative permittivity  $\epsilon_r$ . Close to a charged surface, this approximation can become inaccurate.
- **No specific ion–surface interactions:** The model neglects chemical or specific interactions between ions and the metal surface.

Thus, the main limitations can be summarized as:

|                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| point-ion approximation,<br>no finite ion size,<br>no maximum near-surface concentration,<br>no finite distance of closest approach,<br>constant continuum permittivity,<br>no specific ion–surface interactions,<br>no ion–ion correlations. |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### 3.2) Strengths of the Gouy–Chapman model

The main strengths of the Gouy–Chapman model are:

- **Analytical simplicity:** The model gives closed-form expressions for the potential profile, ion concentration profiles, surface charge density, and double-layer capacitance.

- **Clear physical picture:** It describes how the electrostatic potential decays from the charged surface into the bulk electrolyte.
- **Diffuse-layer screening:** The model captures the basic idea that counter-ions accumulate near the charged surface while co-ions are depleted.
- **Debye length:** It introduces the Debye length,

$$\lambda_D = \left( \frac{\varepsilon_0 \varepsilon_r k_B T}{2e_0^2 N_A c_b} \right)^{1/2}, \quad (47)$$

which gives the characteristic screening length of the electrolyte.

- **Concentration-dependent capacitance:** It predicts that increasing the bulk electrolyte concentration decreases the Debye length and increases the diffuse-layer capacitance:

$$C_{\text{dl}}^{(\text{GC})}(0) = \varepsilon_0 \varepsilon_r \lambda_D^{-1}. \quad (48)$$

- **Useful reference model:** It serves as a simple baseline for more advanced models that include additional physical effects, such as finite ion size, solvent response, and specific ion–surface interactions<sup>a</sup>.

Thus, the main strengths can be summarized as:

analytical and mathematically simple,  
 captures diffuse-layer screening,  
 predicts counter-ion accumulation and co-ion depletion,  
 introduces the Debye length as a natural length scale,  
 predicts the concentration dependence of capacitance,  
 serves as a useful reference model for more advanced theories.

In summary,

The Gouy–Chapman model is qualitatively useful,  
 but quantitatively incomplete for real electrochemical interfaces.

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<sup>a</sup>This is an extra detail by Octavius to make the transition between exercise lectures smoother. Models such as Gouy–Chapman–Stern, finite-size ion models, dielectric saturation models, and specific adsorption models were not introduced in Exercise 2.

## References

- [1] O. Schmidt, “Exercise 2: Thermodynamics of electrochemical cells,” 2026. Problem sheet.